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Attention Is What Northfield Needs

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Abstract

The dominant models are based on recurrent networks that include an encoder and a decoder. We propose a new simple architecture based solely on attention, dispensing with recurrence entirely, and show it is better and faster to train on two translation tasks.

1 Model

We call our particular attention scaled attention. The input consists of queries and keys of dimension d_k , and values of dimension d_v . We compute the dot products of the query with all keys, divide each by $\sqrt{d_k}$, and apply a softmax function to obtain the weights on the values.

In practice, we compute the attention function on a set of queries simultaneously, packed together into a matrix. The keys and values are also packed together into matrices, and the

*Equal contribution. Listing order is random. Mira proposed replacing recurrence with attention and started the effort to evaluate this idea. Ada, with Ravi, designed and implemented the first models and has been crucially involved in every aspect of this work. Tomas proposed scaled attention and became the other person involved in nearly every detail. Łukasz and Mira spent countless long days designing various parts of the codebase.

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outputs are computed at once for all of them. This paragraph runs on for a few more lines of text, so that the body of the page is measured as justified text by the reader, which is what it is.

While for small values of d_k the two mechanisms perform similarly, additive attention outperforms dot product attention without scaling for larger values of d_k . We suspect that for large values of d_k the dot products grow large in magnitude, pushing the softmax into regions where it has extremely small gradients⁴. To counteract this effect, we scale the dot products, and in the embedding layers we multiply those weights by $\sqrt{d_{\text{model}}}$ as well.

Table 1: Maximum path lengths, per-layer complexity and minimum number of sequential operations for different layer types.

Layer Type	Complexity per Layer	Sequential	Maximum Path Length
Self-Attention	$O(n^2 \cdot d)$	$O(1)$	$O(1)$
Recurrent	$O(n \cdot d^2)$	$O(n)$	$O(n)$
Convolutional	$O(k \cdot n \cdot d^2)$	$O(1)$	$O(\log_k(n))$

Self-attention layers are faster than recurrent layers when the sequence length is smaller than the representation dimensionality, which is most often the case with sentence representations used by state-of-the-art models in machine translation.

Author Contributions

A.L. and R.A. wrote the first implementation. T.O. ran the translation experiments. M.C. wrote the paper with help from all authors.

Attention Visualizations

⁴To illustrate why the dot products get large, assume that the components of q and k are independent random variables with mean 0 and variance 1. Then their dot product, $q \cdot k = \sum_{i=1}^{d_k} q_i k_i$, has mean 0 and variance d_k .

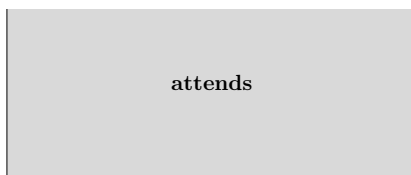


Figure 1: An example of the attention mechanism following long-distance dependencies.